

ULTRON 2.7 Phase 5 Execution Report

Interactive prototype and reusable assistant runtime

Project	ULTRON 2.7
Phase	Phase 5: Interactive Prototype
Workspace	Local workspace checkout: U2.7
Status	Implemented and verified

Executive Summary

Phase 5 completes the basic working prototype by adding an interactive assistant console on top of the existing ULTRON safety pipeline. The new console keeps the assistant text-first while making repeated commands, confirmation flows, and JSON inspection practical for demos and iterative testing.

Phase 5 Scope

- Create a reusable runtime object for repeated assistant commands.
- Add an interactive console entry point with concise human-readable responses.
- Keep the same planner, validator, policy, executor, and audit path for one-shot and interactive usage.
- Add confirmation, JSON inspection, help, and exit commands inside the console.
- Document the final prototype workflow and provide a Windows convenience launcher.

What Was Implemented

Area	File(s)	Result
Reusable runtime	src/ultron27/runtime.py	RuntimeSettings and UltronAssistant for repeated commands without reloading the dataset.
Interactive console	src/ultron27/console.py	/help, /json, /yes, /exit, and concise text response formatting.
CLI integration	src/ultron27/cli.py	--interactive/-i and --text while preserving JSON as the default.
Launcher	scripts/run_ultron_console.ps1	PowerShell helper to start the console from the repo root.
Docs	README.md, docs/phase5_interactive_prototype.md	Interactive prototype workflow and command examples.

Area	File(s)	Result
Tests	tests/test_pipeline.py	Expanded to 24 regression tests.

Console Commands

Command	Purpose
/help	Show available console commands.
/json	Run a command and print the full JSON payload.
/yes	Confirm a command that requires confirmation.
/exit	Leave the interactive console.

Verification Results

Check	Command	Result
Unit tests	python -m unittest discover -s tests	24 tests passed.
Dataset evaluation	python scripts/evaluate_dataset.py --split test	Intent 1.0, tool 1.0, argument exact match 0.9614, confirmation 1.0.
Safety regression	python scripts/evaluate_safety_regression.py	362 cases, tool accuracy 1.0, policy action accuracy 1.0.
Text smoke test	python -m ultron27 ... --text --no-audit	High-risk command returned confirmation_required with /yes guidance.

Prototype Status

- The project now has a working text-first assistant loop.
- The default behavior remains safe dry-run execution.
- Low-risk actions can be executed through the allowlisted executor when --execute is used.
- Destructive actions remain intentionally unimplemented in the MVP.
- The next major layer should be voice input and voice output on top of this stable console runtime.